P9 — Error log of PROJECT spacetimes

Provenance: every error made during the work, how it was found, and how it was corrected

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Purpose

Hard Rule 7 of the task specification forbids shortcuts and demands that problems be explained. This file therefore records every error that was made while the project was carried out, including errors that were corrected before anything was published. For each error it states what was wrong, how it was detected, what was done, and where the corrected statement can be checked. An error that is not listed here and is found later is to be added, not silently repaired.

Errors in the construction and verification of the Dirac matrices

No.	What was wrong	How it was found	Correction	Where to check
E1	The first version of check `71-P4` stated the block form of the complex projectors of 7+1 with the wrong sign of the off-diagonal blocks.	The program compares the statement with the computed matrices; the check failed.	The statement was corrected: χ_{1671} has the blocks $(0, -i;\ i, 0)$, and $2P^{L}_{1671}$ has the blocks $(1, i;\ -i, 1)$.	`data/matrix_checks.txt`, line `71-P4`
E2	The 7+1 matrices of the project were first defined as the covariant matrices `Γ17[swap04(A)]` of `7+1.nb`. The Dirac equation of the notebook, however, is written with the contravariant matrices `Γ17up`.	Re-reading sections 5.3 and 7.1 of `7+1.nb` while writing the Dirac equation of dirac1671.	The project's matrices are $\gamma^{A}_{1671} =$ `Γ17up[swap04(A)]`; then ω_{1671} equals `Γ17[8]` exactly and χ_{1671} equals `χ17`.	`data/matrix_checks.txt`, lines `71-N2`, `71-V1`, `71-P1`
E3	Check `71-Q9` first claimed that the two-particle content of $\psi^{T} C^{-}_{1671} \gamma^{j}_{1671} \psi$ is $+\tau^{j}_{1671}$.	The check failed.	The content is $-\tau^{j}_{1671}$.	line `71-Q9`
E4	Check `71-P5` first claimed that the unitary matrix U^{c}_{1671} maps P^{L}_{1671} to P^{L}_{1644}.	The check failed.	It maps P^{L}_{1671} to P^{R}_{1644} and P^{R}_{1671} to P^{L}_{1644}; the reason (two different sign conventions for the chirality matrix in `7+1.nb`) is stated in the check and in Sec. 2.6 of "dirac matrices".	line `71-P5`

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E5	A commit was created although one check (E4) still failed, because the commands were chained in a way that did not stop on failure; its message claimed that all checks pass.	Reading the output after the commit.	The check was corrected, the program was run again (all checks passed), and the commit was amended before it left the workspace; it was never pushed in the faulty state. Since then a commit is made only if the check program returns success.	`git log`: commit "Dirac equations (dirac1644, dirac1671) and Pauli classification of fermion pairs"
E6	The first implementation of the exact rank (fraction-free elimination with 128-bit integers) overflowed for 256×256 matrices.	The Rust program is compiled with overflow checks and stopped with an error instead of returning a wrong number.	Ranks are computed modulo the prime $2^{61}-1$, with the argument (in the documentation of `rank`) that a rank found modulo a prime is a lower bound of the rank over the rational numbers, so that a full rank is proved; statements about a null space are in addition backed by an explicit null vector. The Python script confirms the same statements by other methods (Gram matrices, enumeration, exact rational elimination).	`code/spacetimes-rs/src/imat.rs`; `data/matrix_checks_independent.txt`
E7	**Symbol collision introduced by this project.** Until November 12, 2026 the program and the drafts used the names `g44`, `g71`, `eta44`, `eta71`, `gamma44`, `gamma71`, `chi71` and the like for the project's matrices. The author's notebooks use these very names for other objects: `g44` and `g71` are metric tensors of the gravity notebooks, `η44` is a 4×4 metric in `Dirac_Nash_Identity.nb`, and `η71`, `Γ71`, `χ71`, `PL71`, `PR71` of `7+1.nb` differ from the project's 7+1 metric, matrices, chirality matrix and projectors. This is exactly the kind of ambiguity that Hard Rule 5 forbids.	A systematic search of all notebooks for names carrying the tags 44, 71 and 17, made while the symbol table of "dirac matrices" was written.	All project symbols were renamed with the tags 1644 and 1671, which occur in no original file (source code, check statements, data files, documents). Other letters with two meanings were separated as well: the 8×8 matrix σ became σ_{88}, the permutation "π" became swap_{04}, the 8×8 blocks "ρ" of 7+1 became τ^{A}_{1671}, the multi-index "K" was replaced by explicit index sets, and rotation parameters are written θ_{AB} instead of ω_{AB}.	MASTER provenance P1; Sec. 0.2 of "dirac matrices"
E8	In the first draft of "dirac matrices" the proof of Lorentz covariance used the transformation $\psi'(x) = S^{-1}\psi(\Lambda x)$ and contracted the indices of Λ incorrectly.	Re-deriving every formula of the draft before the document was assembled.	The correct statement is $\psi'(x) = S\,\psi(\Lambda x)$ with $S\gamma^{A}S^{-1} = \Lambda^{A}{}_{B}\gamma^{B}$; the derivation is given in Sec. 1.8.2. The faulty draft was never committed.	"dirac matrices", Sec. 1.8.2
E9	The first version of the routine of the Python script that counts invariant pair matrices mixed the linear equations of different generators.	Review of the code before its first complete run.	Each generator contributes its own 64 equations for the 28 unknowns. The corrected routine reproduces the numbers of the Rust program (0 and 10 for 7+1; 6, 10, 1 and 0 for 4+4). The faulty version was never committed.	`code/scripts/verify_matrices_independent.py`, function `invariant_dimension`

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E10	The draft of Sec. 1.6 said that cell 166 of `4+4.nb` verifies all five projector relations; the cell verifies three of them.	Comparing the sentence with the cell.	The text now says which three relations the notebook verifies; all five are verified by checks `44-P1`, `44-P2`. Never committed in the faulty form.	“dirac matrices”, Sec. 1.6